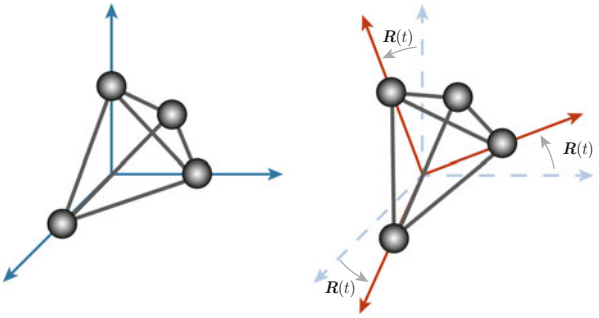


Fig. 4.7 A rigid rotation can be viewed as premultiplication by an orthogonal matrix $\boldsymbol{\Theta}$. In this way the orientation of the position vectors of points in the rigid body at any time t are described by a trajectory in the space of 3×3 orthogonal matrices, $\boldsymbol{\Theta}(t), t \in \mathbb{R}$

$\boldsymbol{\delta}_i(t) = \boldsymbol{\Theta}(t)\boldsymbol{\delta}_i(0)$, see Fig. 4.7, thus we have

$$K_{\text{rot}} = \frac{1}{2}\sum_{i=1}^{\text{N}} m_i \boldsymbol{\delta}_i(0)^T \dot{\boldsymbol{\Theta}}^T \dot{\boldsymbol{\Theta}} \boldsymbol{\delta}_i(0).$$

This expression may be rewritten as

$$K_{\text{rot}} = \frac{1}{2}\text{Tr}\left(\dot{\boldsymbol{\Theta}}\boldsymbol{R}\dot{\boldsymbol{\Theta}}^T\right),$$

where the fixed matrix $\boldsymbol{R}$ is given by

$$\boldsymbol{R} = \sum_{i=1}^{\text{N}} m_i \boldsymbol{\delta}_i(0)\boldsymbol{\delta}_i(0)^T.$$

By choosing an appropriate coordinate frame, it is possible to insure that the symmetric matrix $\boldsymbol{R}$ is a diagonal matrix.

An alternative form for the rotational kinetic energy may be obtained by writing

$$\begin{aligned}\dot{\boldsymbol{\delta}}_i &= \dot{\boldsymbol{\Theta}}\boldsymbol{\delta}_i(0) \\ &= \dot{\boldsymbol{\Theta}}\boldsymbol{\Theta}^T\boldsymbol{\Theta}\boldsymbol{\delta}_i(0) = \dot{\boldsymbol{\Theta}}\boldsymbol{\Theta}^T\dot{\boldsymbol{\delta}}_i,\end{aligned}$$

where we have used the fact that $\boldsymbol{\Theta}$ is orthogonal. Moreover, since $\boldsymbol{\Theta}\boldsymbol{\Theta}^T = \boldsymbol{I}$, we have

$$\dot{\boldsymbol{\Theta}}\boldsymbol{\Theta}^T + \boldsymbol{\Theta}\dot{\boldsymbol{\Theta}}^T = 0,$$